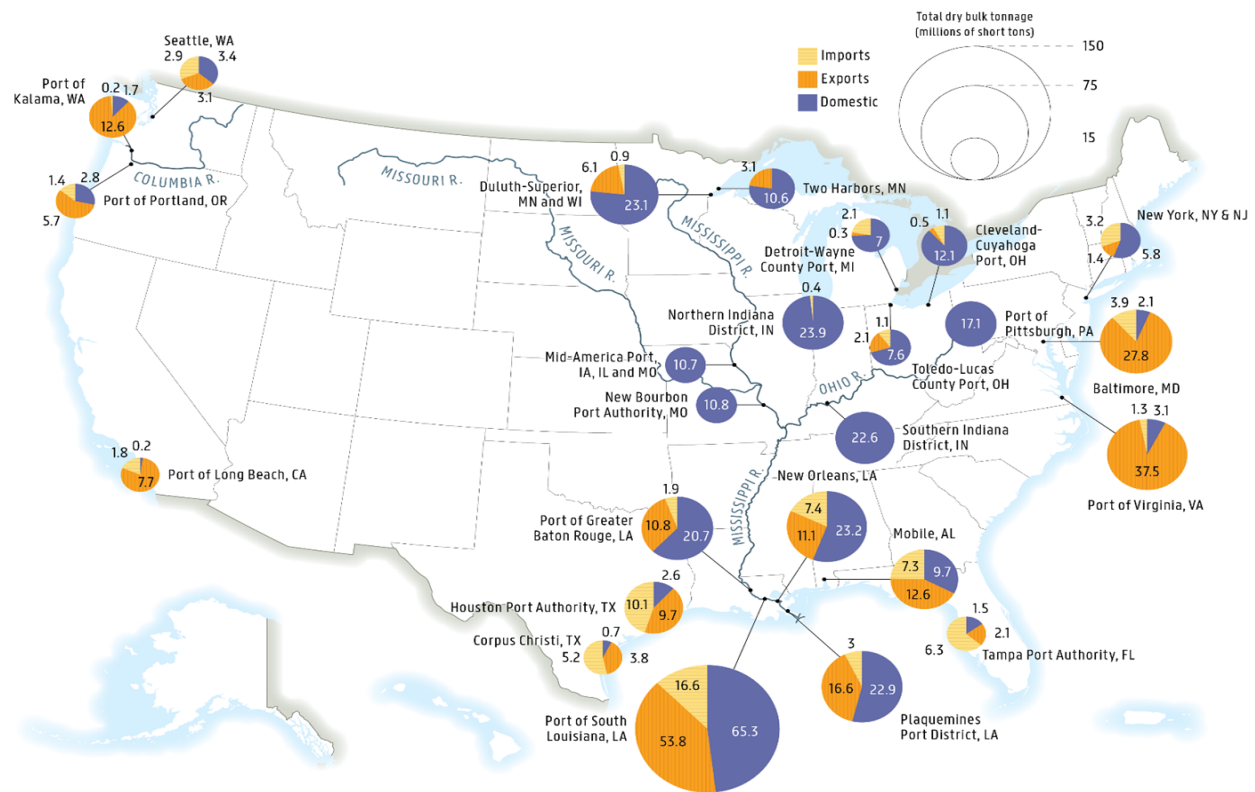




1. Background

This figure is a U.S. map displaying dry bulk cargo tonnage at major ports. Each port is represented by a circular multi-slice chart showing imports (yellow), exports (orange), and domestic shipments (blue). The size of each circle corresponds to total dry bulk volume in millions of short tons.

2. Text Contrast

Port names and numeric tonnage values appear in black text against a white or light geographic background, providing strong contrast. However, tonal variation on the map (rivers, state lines) may slightly reduce clarity in some areas. Numeric labels placed directly on colored slices may experience reduced contrast, particularly on yellow or orange fill areas.

3. Graphic (Slice) Contrast

Color contrast across slices is generally strong: yellow, orange, and dark blue are distinct. However, yellow and orange remain adjacent on the color spectrum and may not be easily distinguishable for users with color-vision deficiencies. Blue slices provide strong visual separation due to their darker tone. Thin strokes around slices help with boundary definition but do not fully mitigate color similarity issues.

4. Color-Blind Safety

Yellow and orange can appear nearly identical for users with red-green color blindness (deuteranopia or protanopia). The dark blue slice remains highly distinguishable. Because the chart relies solely on color to encode import/export/domestic distinctions (no patterns or textures), accessibility may be reduced for color-blind viewers. Additional non-color encoding—patterns, borders, labels—would improve accessibility.

5. OCR / Text Size

Most port labels and numeric tonnage values are small but mostly legible. Ports located in dense regions such as the Gulf Coast and Great Lakes may experience overlapping labels, which limits OCR accuracy. Text embedded inside small pie slices may be too small for OCR to extract reliably.

6. Blur & DPI

The overall image appears clear with sufficient DPI for digital use. Thin lines (rivers, state borders, chart outlines) remain crisp. Fine details such as the smallest circles or labels may degrade slightly if printed or compressed. Maintaining ≥ 300 DPI is recommended for publication-quality accessibility.

7. Alt Text Recommendation

A recommended 508-compliant alt text description: “ Map of the United States showing major dry bulk cargo ports using proportional pie charts to indicate imports (yellow), exports (orange), and domestic shipments (blue). Larger volumes appear at ports such as South Louisiana, Virginia, New Orleans, and Toledo-Lucas County. Smaller ports across the West Coast, Gulf Coast, Great Lakes, and East Coast are also shown with scaled charts. Legend indicates relative tonnage in millions of short tons. ”